Voiland Food Pantry

Final Report Draft 1

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CptS 423 Software Design Project II

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Link to Video: https://youtu.be/85BKI1TZN80

Link to GitHub: https://github.com/WSU-JB5757/WSU_Marketing_423

I. Introduction

As stated in the promotional material*, Washington State University, in an effort to reduce food insecurity, has opened food pantries on campus to provide free food. The development team has created a program to read card data for volunteers and beneficiaries, and provide a database to track the pantry inventory. In addition to the Voiland Food Pantry, the program has potential uses at other campus facilities. The sponsor has requested a rollout plan including all necessary technical documents, testing and specifications of the current program to prepare it for scaling, maintenance, distribution and future revisions by other facilities and development teams. The software program will be tested, documented and prepared for deployment with documentation and revisions from the product development team. The marketing team does not provide code or product development for future program revisions. Instead, it organizes information and plans goals for product rollout and future development.

II. Team Members & Bios

II.1. Jodie Butterworth

Jodie is a senior Computer Science major at Washington State University. Her skills include UI/UX mobile and web development, graphic design, and technical writing. She has prior experience with web development and has worked with WordPress, as well as programming languages such as Python, C/C++/C#, HTML, and CSS. For this project, her responsibilities include front-end development and UI design.

As described in earlier Project Drafts**.

* Introduction from the promotional material abstract for the poster presentation.

** Bio from the team bio section of the development team project report dated 12-7-25.

III.Project Requirements Specification

As agreed in the Client meeting, the scope of the rollout for the marketing project is:***

- Create timeline for product deployment with multiple locations and stakeholders.
- Write a technical document for users to edit program.
- Write technical document for users to troubleshoot common problems.
- Create a plan for continued maintenance including hand-off to future project teams.
- Plan marketing meeting with potential future clients.
- Write the technical specifications and hardware requirements for the project.
- Identify potential costs and fees of operating the system.
- Identify quality and performance issues with testing, implement repairs, and record the process and results.
- Create a user's guide for practical daily operation.

III.1. Project Stakeholders

List of Stakeholders:

- Clients for the Voiland Food Pantry who want a rollout plan for the scanning software to manage inventory and card readers to manage client and volunteer data
- Potential clients in other departments who want to improve their inventory, volunteer and employee data management
- Future users, team project developers and administrators who want clear documentation of the program and how to use and troubleshoot it
- Current development team who want to develop the project with minimal issues.

*** As described the meeting minutes for the meeting on 3-5-26

III.2. Use Cases

- U1 As a volunteer of the Voiland Food Pantry, I want to read the user manual so that I know how to run the terminal.
- U2 As an admin of the Voiland Food Pantry, I want to read the admin and maintenance manual so that I know how to change the program title.
- U3 As a developer on a future improvement to the program, I want to look at the technical specs so that I know the hardware that is needed.
- U4 As a new client I want to read the troubleshooting manual so that I know how to operate the database.
- U5 As a developer of a future improvement to the program, I want to read the testing documentation so that I understand the program operation and the testing used.

III.3. Functional Requirements

Based on the objectives from a client meeting:***

- FR1 Development team to produce user guide for daily operations.
- FR2 Development to create a technical document for editing and maintenance by future admins and project teams.
- FR2 Development team to write a report on any costs or fees in operating the product and any foreseen costs of fees of future operations.
- FR3 Development team to produce technical document for technical specifications.
- FR4 Development team to make a technical document for troubleshooting common problems.
- FR5 Development team to record procedures and results of testing processes and the repairs implemented.

III.4. Non-Functional Requirements

Based on the discussion from a client meeting:***

- NR1 Improve communications with development team.
- NR2 Teams to be used in timeline planning.
- NR3 Meetings to be held with potential future clients.
- NR4 Client to be CC in all email communications for the project.
- NR5 Documents to be edited and easy to use.
-

III.5. Standards

This project is related to a software design project, but is itself not software based. Software testing will be done and compared with other testing results. Teams will be used as project management and communication software. Email is the main communication method.

I am planning to implement AI-based project management to create a more complete and interactive rollout planner. This is still in the planning phases as I didn't think of it until recently. Teams and email will still be used as communication. GitHub will still be used as storage and to turn things in. This will not be a large program, probably made with Zapier or a similar AI site.

Aside from that, the rollout documentation mush be edited, complete, easy to use and contain all the required information.

IV. Software Design - From Solution Approach

For the rollout portion, this section does not apply, other than testing results comparisons.

For the AI portion, it will be a better timeline organizer with more features for communication and AI assistance incorporated. It will not be a large program, more an experiment that will require testing and revision to ensure correctness in the results. It will rely on external sites.

IV.1. Architecture Design

Not applicable.

IV.2. Data design

Not applicable.

IV.3. User Interface Design

Not applicable.

IV.4. Constraints

See the project scope under Project Requirements.

V. Test Case Specifications and Results

V.1. Testing Overview

The terminal has been installed, but I have not completed testing on it. I plan to do testing after the break.

V.2. Environment Requirements

The technical specs should describe the required hardware.

V.3. Test Results

The terminal has been installed, but I have not completed testing on it.

VI. Projects and Tools used

I'm using GitHub, Teams and word processors.

VII. Description of Final Prototype

The final project will include a user manual, a manual for admins and maintenance, the technical specs of the program, a troubleshooting manual (which could be included with user manuals), an analysis of operating costs, testing and testing records, a timeline with tasks for the rollout, and I want to add, an AI planner.

VIII. Social Responsibility and Broader Impacts

The food pantry provides a valuable community service. Improving their software to more efficiently track volunteer hours, customer use, and product inventory will enable them to do their job more efficiently. A rollout plan makes the transition for the next team who takes over with the project improves easier for them. It also connects with potential clients and provides both current and future clients and administrators the records and instructions they need to better utilize the program.

IX. Product Delivery Status

The last item delivered was the booklet for the poster presentation. It out that it wasn't needed, but no one informed me of that, so I sent it to the client by email.

The last meeting established the scope of the project and the desired deliverables. The development team sent a simple user guide for testing purposes. The user guide was forwarded to the client for review. A timeline has also been established on Teams, as per client request.

X. Conclusions and Future Work

X.1. Limitations and Recommendations

I do not directly contribute code for the development team. They are responsible for creating the needed documents to specifications so that they will be useful.

X.2. Future Work

After writing all this, I will complete the tasks that the client requests, but I have no control over the quality or quantity of the rollout documentation produced by the team.

I think that is why the idea of making an AI rollout planner appeals so much to me. It is a useful improvement to the marketing rollout project that does not rely on the development department creating guides and documents.

XI. Acknowledgements

I thank Maynard Poleng Siev for sponsoring the marketing project. I thank the development team and my instructor for reviewing the marketing project and for putting in the extra work to make it viable.